Paleomagnetic result of the Lower-Cretaceous from Luanping basin, Hebei Province and its tectonic implications*

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Abstract A paleomagnetic study of about 95 samples from 16 sites sampled in the Early Cretaceous in Luanping basin in Hebei Province was reported. Stepwise thermal demagnetization was used to isolate magnetic components. Most samples have a characteristic direction with a high temperature component above 500℃. The tectonic-corrected data are $D=347.8°$, $I=50.4°$, $\alpha_{95}=7.1°$, and the corresponding pole position is at 76.1°N, 346.3°E, with $dp=6.4°$, $dm=3.8°$, paleolatitude $\lambda=31.1°$N. This result indicates a counterclockwise post-Cretaceous rotation of 30.7° ±9.8° with respect to the stable Ordos basin in the west of North China Block, and a non-significant northward motion. This rotation could be related to local fault action or structural detachment, or regional NNW-NWWward motion and collision of Kula-Pacific plate with eastern China since the Early Cretaceous.

Keywords: Luanping basin, Early Cretaceous, counterclockwise rotation, Kula-Pacific plate.

Since the 1980s, a large number of paleomagnetic studies have been carried out in the Ordos basin in the west of North China Block (NCB)[1—5]. A number of reliable Jurassic-Cretaceous paleomagnetic data have been obtained to establish the apparent polar wander path (APWP) for the whole NCB, and to provide important fundamental information for studying its structural evolution. A lot of geological studies have shown the distinct tectonic differential movement between the eastern and western part of NCB since the Mesozoic. The western part of NCB is characterized by large Jurassic-Cretaceous fault basin, while the eastern part is characterized by small Middle Jurassic-Cretaceous fault basin, large scale magmatic explosion and intrusion, and severe structural activity. So, paleomagnetic studies may contribute to a better understanding of the Mesozoic evolution of NCB by correlating the Jurassic-Cretaceous paleomagnetic data with that from other tectonic units. A considerable amount of paleomagnetic studies have been carried out in the Jurassic-Cretaceous sediments in the eastern part of NCB. It is difficult to interpret the paleomagnetic data because of the complex structure, multi-periodic volcanic activity and widespread remagnetization related to the Yanshannian orogeny in this region. In this paper we report the paleomagnetic result obtained from lower Cretaceous sediments in Luanping basin, and discuss its tectonic implications by comparison with paleomagnetic results from adjacent blocks.

1 Sampling and paleomagnetic measurements

The Luanping basin is one of the regions in NCB in which the late Jurassic—early Cretaceous volcanic sediments are well developed. The Luanping group can be divided into two subgroups.

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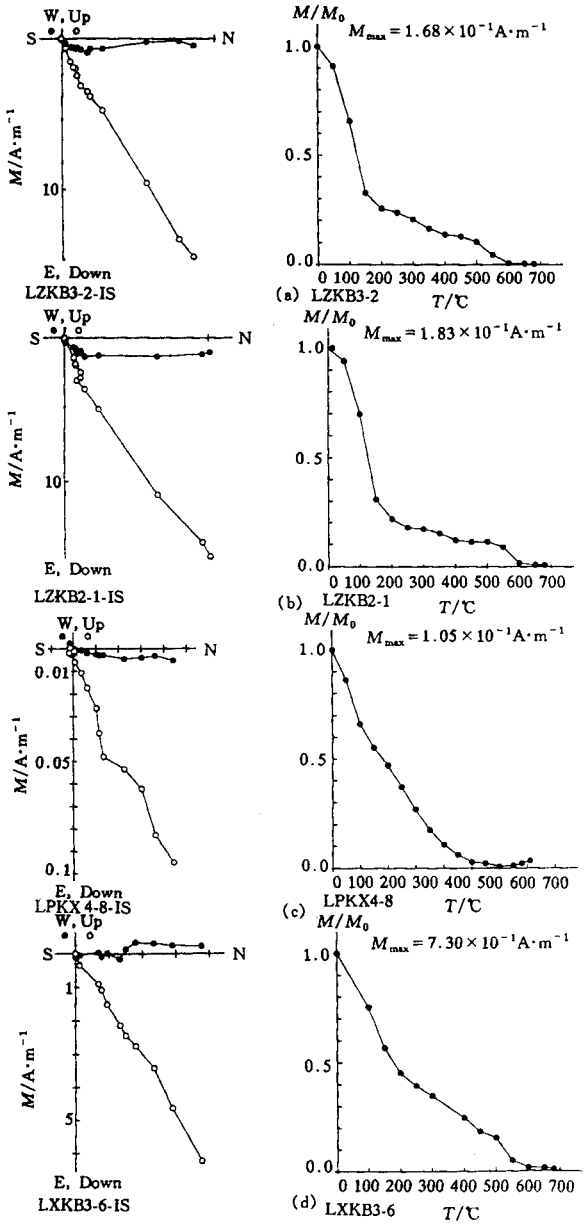


Fig. 1. Orthogonal vector diagrams and normalized decay curve of the magnetization for representative specimens. Solid/open symbols refer to projection on the horizontal/vertical plane in geographic coordinates. (a, b): double-component specimen; (c, d): single component specimen.

The lower subgroup (Donglingtai group) including Baiqi and Zhangjiakou formations is composed of intermediate-acidic volcanic sediments, and the upper subgroup (Rehe group) is composed mainly of lacustrine sediments bearing famous Rehe biota. The sediment series consists of five formations. The age of this stratum which belongs to the Late Jurassic or Early Cretaceous has been a controversial topic[6—9]. Recent studies[10,11] show that the upper and lower subgroups of Luanping group may belong to the Lower Cretaceous though there is some difference in the isotopic ages[12—15]. This question still needs further work.

A total of 16 sedimentary sites (more than 120 cores) were sampled with a portable drill. Bedding attitudes are monocline in this region.

All samples were measured and demagnetized at the paleomagnetic laboratory of the Chinese Academy of Geological Science. Each sample was demagnetized by heating.

Typical examples of orthogonal plots obtained by this procedure are shown in fig. 1, progressive thermal demagnetization of samples classified the sampling sites into two categories. In one category, all samples exhibited two components of magnetization (fig. 1(a), (b)), a low unblocking temperature component around 150℃ and a high-temperature component from 500℃ to 580℃. The first component may be carried by titano-magnetite ($x = 0.6$)[16], and the second high-temperature component is carried by magnetite and is primary. In the other category, all samples exhibited a single-component magnetization (fig. 1 (c), (d)). The unblocking temperature ranges from 500 to 600℃, indicating that this component is carried by magnetite. Generally, the lower temperature component is close to the present Earth Field (PEF) direction. In order to determine the nature of the minerals carrying the remnant magnetization, petrographic analysis and electric probe (tables 1, 2) have been carried out on some typical rocks. The results show that the magnetic minerals are primary titano-magnetite and magnetite. So, we consider that the high temperature component is primary.

Table 1 Electronic probe results of chemical composition for representative specimens from Cretaceous formation in Luanping basin

Specimen \ Composition	FeO	MnO	Cr_2O_3	TiO_2	CaO	K_2O	SiO_2	Al_2O_3	MgO	Na_2O	Total
Lz-1	69.86	0.16	0.31	25.48	0.42	0	0.5	1.07	0.77	0.03	98.72
Lz-2	81.63	1.81	0.09	8.14	0.03	0	0.27	1.58	0.91	0.1	94.56

Table 2 Computed results based on four oxygen atoms

Specimen \ Composition	Fe	Mn	Cr	Ti	Ca	K	Si	Al	Mg	Na	Mineral
Lz-1	2.32	0.05	0.01	0.76	0.02	0	0	0.05	0.05	0	titano-magnetite
Lz-2	3.14	0.07	0	0.28	0	0	0.01	0.01	0.06	0.01	magnetite

The paleomagnetic direction of each sample was analyzed by principal component analysis[17], and the characteristic directions calculated for each site using Fisher's statistics[18] are shown in table 3.

Table 3 Characteristic remnant magnetization directions for Lower Cretaceous of Luanping basin (117.5°E, 41.0°N)

Site	N	D_g/(°)	I_g/(°)	D_s/(°)	I_s/(°)	κ	α_{95}/(°)
LPKB1	6	43.6	64.9	349.5	53.2	61.3	8.6
LPKB2	11	24.0	60.1	347.1	43.8	42.2	7.1
LPKB3	7	19.0	64.5	340.6	44.6	160.1	4.8
LPKB4	7	329.6	80.2	311.2	48.9	105.2	5.9
LPKB5	3	339.6	61.6	323.7	32.6	693.4	4.7
LPKZ1	8	60.5	47.7	14.4	53.6	651.1	2.2
LPKZ3	9	320.9	68.8	354.8	59.9	277.7	12.4
LPKZ4	7	345.5	54.1	0.3	41.7	114.9	8.8
LPKZ5	5	206.0	−75.9	207.7	−64.9	600.1	9.0
LPKX1	3	5.1	61.6	357.8	42.4	148.7	10.1
LPKX2	3	6.9	71.8	356.0	52.5	73.7	14.5
LPKX3	8	312.4	60.3	323.9	42.4	25.3	11.2
LPKX4	6	1.1	66.5	348.1	43.7	59.6	8.7
LPKX5	4	358.8	67.0	352.8	49.4	350.1	4.9
LPKJ1	8	336.2	70.6	357.1	61.3	693.4	3.5
Average	15(95)	3.0	68.2	347.8	50.4	29.7	7.1
Pole:	long. = 346.3°E,	lat. = 76.1°N,	dp = 6.4°,	dm = 9.5°,		paleolat. = 31.1°N.	

N, Number of specimens; D, declination; I, inclination; subscripts g and s: geographic and stratigraphic coordinates, respectively; κ, precision parameter; α_{95}, 95% confidence angle.

The mean declinations and inclinations calculated before and after bedding correction for each site are given in table 3. The statistical results show that the dispersion of the ChRM direction does not show much change after tilt correction ($k_g/k_s = 28.3/29.7$). This indicates that the fold test is insignificant. Of the 16 sites, only one site has reversed polarity, and its direction is not completely antipodal to the other normal ones.

Although the above results indicate that the mean direction did not pass the fold and reversed tests, considering that the mean high-temperature component is statistically indistinguishable from the PEF, and the results of petrographic analysis and electric probe, we believe that the high temperature component carried by magnetite may represent the primary stable component of the magnetization.

The mean direction converted to the virtual geomagnetic pole (VGP) for the early Cretaceous

of the Luanping area gives 346.3°E, 76.1°N, $dp = 6.4°$, $dm = 9.5°$.

The lack of significant fold test of 95% level (there is just a small increase in the precision parameter) does not allow us to ascertain the primary nature of the lower Cretaceous magnetization. However, one can see that the mean direction is significantly different from that of recent geomagnetic field.

The paleolatitude and declination of Ordos basin calculated at our reference site (Luanping basin) are found to be, $\lambda = 39.2°$N, $D = 18.5°$. There is a small, insignificant difference in paleolatitude between Ordos basin and Luanping basin.

2 Result and discussion

This rotation is in good agreement with the geological evolution. To assess the tectonic significance of the paleomagnetic results we have considered the paleomagnetic data from adjacent blocks (table 4). Comparison of these pole positions shown in fig. 2, with circles of 95% confidence, indicates that at the 95% confidence level no significant differences can be observed among NCB, SCB, Mongolia fold belts and central Asia during the (Middle) Late Jurassic. This shows that they have collided since at least the (Middle) Late Jurassic, but these (Middle) Late Jurassic poles differ significantly in paleolatitude from that from Eurasia plate (as far as 20°), and the paleolatitude gap in the (Middle) Late Jurassic disappeared in the Early Cretaceous. This may imply that an important tectonic event, such as intercontinental deformation (involving some 2 000 km of shortening), has occurred in post Late Jurassic time. This conclusion is in good agreement with geological evidence[19,20]. The Jurassic thrust sheets with minimum displacement of 120—180 km have been discovered within the "Hercynian-Indosinian" orogenic belt of the Beishan of China and Gebi area. The thrusts strike E-W, and extend over 1 200 km in length. The thrust sheets formed during the (Late) Middle Jurassic age of 155.1 ± 10 Ma BP. Though the nappe displacement distance is smaller than the northward shortening distance, the time of the thrust sheets forming is similar to that of the paleolatitude gap disappearing (from Middle-Late Jurassic to Early Cretaceous).

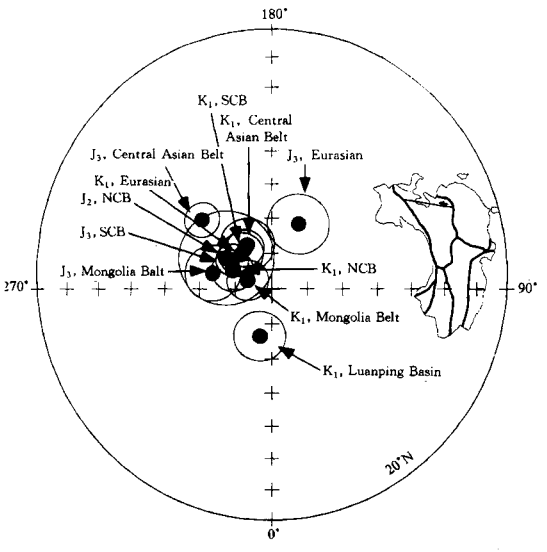


Fig. 2. Comparison of the Late-Jurassic/Early Cretaceous pole position from NCB, SCB, Mongolia fold belt, central Asia, and Eurasia block.

The Early Cretaceous pole from Luanping has similar paleolatitude to the others but has an evident declination gap, which indicates that the Luanping basin has been rotated counterclockwise through 30.7° ± 9.8° with respect to the above blocks since the Early Cretaceous. The amount of rotation R is given by $R = D_{obs} - D_{exp}$, where D_{obs} and D_{exp} are the observed and the expected declination at Luanping, respectively. The expected declination value (D_{exp}) is calculated from the VGPs of Ordos basin. The observed value from Luanping basin is $D = 347.8°$, $I = 50.4°$, $\lambda = 31.1°$N at the reference point 117.5°E, 41.0°N. The expected value is $D = 18.5°$,

$I = 58.5°$, $\lambda = 39.2°$N. The amount of rotation (R and its uncertainty (ΔR)) is $30.7° \pm 9.8°$ with respect to Ordos basin, and the paleolatitude gap is $8.1° \pm 9.8°$. Comparison of the observed and expected value of declination and paleolatitude shows that the Luanping basin experienced counterclockwise rotation with insignificant north-south translation since the Early Cretaceous. Such a counterclockwise rotation has also been recognized in SCB[25].

Table 4 Paleomagnetic data of Luanping basin and adjacent regions for Lower Cretaceous

Area	Age	N	Pole position		A_{95}/(°)	Reference
			ϕ/(°E)	λ/(°N)		
Ordos	J_2	6	232.8	74.6	4.5	[2]
SCB	J_3	5	235.3	74.6	13.3	[21]
Mongolia fold belt	J_3(?)	8	254.8	73.0	7.8	[22]
Central Asia	J_3(?)	14	224.6	62.4	4.9	[22]
Eurasia	150 Ma	7	157.8	70.0	6.6	[23]
Ordos	K_1	10	208.7	75.8	7.5	[2]
Luanping basin	K_1	15	346.3	76.1	7.1	this study
SCB	K_1	3	220.6	77.0	6.1	[3]
Mongolia fold belt	K_1	6	249.5	82.9	5.7	[22]
Central Asia	K_1	1	205.3	72.5	7	[24]
Eurasia	120 Ma	7	212.6	76.3	5.2	[23]

N, Number of studies or poles; ϕ/λ, VGP longitude and latitude; A_{95}, VGP 95% confidence angle.

We now compare our results with other published Jurassic-Early Cretaceous results known to us (table 3). We can see from data listed in table 4 not only that the Late Jurassic poles of NCB and SCB are statistically similar but that they are also similar to those of Mongolia and central Asia. In any case, it appears that in the Late Jurassic, the SCB, NCB and the Mongolia block formed a single entity but were still far from being similar.

The orientation of major structural lines and the long axis of volcanic sedimental basin rotate northward with time from east to west[26]. This geological phenomenon was similar to the results expected by paleomagnetic study. What caused the difference in tectonic rotation in the eastern and western part of NCB during Yanshannian episode?

More and more deep geological information shows that the lithosphere of NCB is not a rigid block, and it bears the effect of the plate margin movement[27—29]. In fact, within the NCB, the lithospheric structure is very inhomogeneous, the thermodynamic state also varies largely, and the movement of deep rheosphere differs for various regions. So, the dynamic process in this region is not a single mechanical problem. It should include the factors of physics, chemistry and mechanics. Ordos basin is one of the stable blocks in NCB, with a crustal thickness of 40—60 km, a lithospheric thickness of 106—130 km, a homogeneous distribution of geotherm, whereas, in Yanshan area, the crustal thickness is 34—38 km, and the lithospheric thickness is 85—100 km. On the whole, the lithosphere is thicker in the western part than that in the eastern part, and the thickness of the crust also has the same feature.

Regional geological investigations[30] show that the fault in the west-north margin of Luanping basin is a synsedimentary one, while the Shangyi-Pingquan fault in the south and the Fengning-Longhua fault in the north, which began to move from the Archeozoic to Proterozoic, showed thrusting or even napping with right-lateral shearing. This character may result in the

counterclockwise rotation of Luanping basin. In addition, Zheng[31] suggested that the structural detachment is a style of structural action which may cause a series of blocks to rotate in this region.

Many scholars[25] related the Mesozoic structural movement of NCB and eastern China to that of the Pacific plate. The paleopacific active continental margin of eastern Asia formed in the Middle Jurassic[32—37]. The Kula-Pacific plate underthrust beneath eastern Asian continent in NNW direction in the Middle Jurassic-Early Cretaceous. In the Middle Cretaceous, the Kula plate was consumed, and the Pacific plate continued a NNW compression accompanied by left-lateral slip. From the Early Eogene, the direction of the Pacific plate movement started to change from NNW to NWW, underthrusting beneath eastern Asian continent continually. The above movement of Kula-Pacific plates formed a stress field compressing with NNW-NWW in eastern China since the Middle Jurassic. Wan[38] pointed out that the direction of the maximum principal stress was NWW-SEE in eastern China in Yanshannian stage (208—135 Ma), and the stress field appeared weaker in Ordos and Sichuan basin. In the Sichuanian stage (135—52 Ma) the direction of the maximum principal stress changed to NE-SW, and the stress field appeared stronger in south-western China and weaker in north-eastern China. In the North Sinian (52—23.3 Ma) the direction of principal stress was NW-SEE, which indicates that the paleo-stress field is dominated by NNW-NWW compression in eastern China since the Yanshannian stage. However, in the west part of North China, due to the lithospheric inhomogeneity of North China, the structural stress field formed by Pacific plate underthrusting got weaker from east to west, and the effect of Pacific plate subducting diminished.

As mentioned above, the Luanping basin in Yanshan area yielded a counterclockwise rotation of $30.7° \pm 9.8°$ with respect to Ordos basin since the Early Cretaceous. This rotation may result from local fault action, or structural detachment, or the regional NNW-NWWward motion and collision of Kula-Pacific plates with eastern China since the Early Cretaceous. Then, which factor is the major one? We still do not have sufficient evidence to answer this question. We shall now address the question of whether the whole Yanshan region and the eastern China rotate as a single block. Based on the detailed paleomagnetic study on the Jurassic-Cretaceous sediments in Yanshan region and the eastern China, the difference of Mesozoic structural evolution in geological time and space can be studied further.

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